



Physiological Responses to Agonist–Antagonist Superset Resistance Training

Julius Fink¹ · Brad Jon Schoenfeld² · Mikako Sakamaki-Sunaga³ · Koichi Nakazato³

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Abstract

Purpose To investigate the physiological responses to low-load, superset resistance training (two exercises for the agonist and antagonist muscles performed without rest between exercises) to failure using elastic bands.

Methods Twenty-three athletes were randomized to either a superset group (S, $n = 12$, average age: 19.8 ± 1.5 years) or a traditional set group (T, $n = 11$, average age: 20.1 ± 1.4 years). Strength, cross-sectional area (CSA) and muscular endurance of the biceps and triceps brachii were assessed before and after 8 weeks. Acute responses (muscle thickness) were measured during one testing session.

Results Muscle thickness of the biceps significantly increased in both T group ($P < 0.05$) and S group ($P < 0.05$) after a single bout of Training. The triceps did not show significant increases in either T group ($P > 0.05$) or S group ($P > 0.05$). Blood lactate also increased in both groups after one bout of training (T: from 1.3 ± 0.3 to 5.5 ± 2.4 mmol/L, S: from 1.4 ± 0.5 to 5.1 ± 1.5 mmol/L, $P < 0.05$). After 8-week training, both groups showed significant increases in the biceps (T: $13.2\% \pm 5.0\%$; S: $12.9\% \pm 7.3\%$, $P < 0.05$) and triceps (T: $9.5\% \pm 9.3\%$, S: $4.8\% \pm 4.1\%$, $P < 0.05$) without differences between groups. Increases in one repetition maximum for the bench press ($7.8\% \pm 6.5\%$, $P < 0.05$) and maximal voluntary contraction for the arm extensors ($9.3\% \pm 11.6\%$, $P < 0.05$) were observed for the T group only. Increases in muscular endurance were observed only in the S group for the bench press ($26.0\% \pm 19.1\%$, $P < 0.05$) and the barbell curl ($17.2\% \pm 16.6\%$, $P < 0.05$).

Conclusions Superset training may enhance muscular endurance while attenuating maximal strength gains. There does not appear to be a hypertrophic benefit to performing superset training, but it may provide a time-efficient strategy to achieve adaptations in muscle mass.

Keywords Muscle hypertrophy · Strength · Endurance · Thickness · Cross-sectional area

Introduction

Several longitudinal studies in young populations have shown similar muscle hypertrophy with low load [$\sim 30\%$ 1 repetition maximum (RM)] resistance training (RT) as compared to high load ($\sim 80\%$ 1RM) RT [17, 22, 34] when sets are performed to muscular failure, provided the number of

sets are kept constant. Acutely, there is evidence that low-load RT (30% 1RM) triggers a more prolonged duration of post-exercise muscle protein synthesis as compared to high load (90% 1RM) RT [9]. Although strength seems to increase more in response to high loads ($> 60\%$ 1RM), muscle hypertrophy seems to react similarly to low ($< 60\%$ 1RM) and high loads [33]. Although some studies do report superior increases in muscle mass for high load RT [14, 15], the sets were not carried out to muscle failure, which may attenuate the hypertrophic response of low-load training.

The outcome of a given RT protocol depends on multiple factors such as the contraction modes per repetition, duration of each repetition, rest between repetitions, time under tension, proximity to muscular failure, range of motion, and recovery time [35]. Several methods are widely used among trainees to enhance the hypertrophic response of resistance training (RT) such as drop sets, negative sets,

✉ Julius Fink
j-fink@juntendo.ac.jp

¹ Department of Urology, Graduate School of Medicine, Juntendo University, 2-1-1, Hongo, Bunkyo-ku, Tokyo 113-8421, Japan

² Department of Health Sciences, Lehman College, Bronx, NY, USA

³ Graduate Schools of Health and Sport Science, Nippon Sport Science University, Tokyo, Japan

forced repetitions and supersets [3, 31]. Indeed, the heightened stimulus of these strategies may translate into increased levels of mechanical and metabolic stress, triggering the activation of factors proposed to be involved in anabolism including hormonal elevations [3, 32], muscle swelling [32], muscle fiber recruitment [31] and muscle protein synthesis [8], ultimately leading to increased muscle hypertrophy.

In contrast to traditional training in which the trainee executes all sets for one exercise before moving on to the next exercise, superset training is one of the most popular methods for increasing hypertrophic responses among bodybuilders and other trainees [13]. In this study we refer to superset RT as a combination of two exercises for agonist and antagonist muscles, also called “agonist–antagonist paired set” training [28]. The strategy involves performing two exercises in immediate succession with minimal rest between sets. Even though the term “superset” is widely used in the fitness field, it is a broad definition that does not specify the exact combination of exercises. Indeed, a superset can be performed on the same muscle group using different exercises with the goal to pre-exhaust the muscle and stimulate the muscle from several different angles.

Superset training has been investigated with regard to its effects on power and strength [26–30], intensity and fatigue markers [36], total training volume and perceived exertion [4], and energy expenditure and postexercise oxygen consumption [19]. The results of one long-term study indicate improved strength outcomes for superset RT as compared to traditional training while being more time-efficient [28]. Several metabolic stress markers such as creatine kinase and lactate seem to increase in response to superset training [36]. One study showed that energy expenditure relative to time and excess postexercise oxygen consumption are greater in superset protocols as compared to traditional protocols [19], which could signify that muscular endurance might improve to a greater magnitude with superset training. Other research demonstrated that force output of the agonist can be increased by prior contraction of the antagonist muscle, probably due to increased accumulated elastic energy in the muscle tendon and a decrease in antagonist muscle activity (i.e. reciprocal inhibition) [18, 31]. Coactivation (concurrent activation of agonist and antagonist muscles) might change the pattern of ballistic contractions and lower resistance due to pre-exhaustion of the antagonist, improved activation of the agonist due to mutual innervation [28]. Furthermore, greater metabolic stress and hormonal responses might increase muscle hypertrophy following superset training [36].

Even though several studies have investigated aspects of superset RT, research on the effects of superset RT on muscle hypertrophy is lacking [31]. Previous research comparing energy expenditure in superset vs. traditional RT (single sets separated by 60 s rest intervals) showed increased exercise

energy expenditure, blood lactate and excess post-exercise oxygen consumption, pointing to elevated metabolic stress and fatigue in superset RT [19]. Indeed, larger metabolic stress experienced with improved training density strongly relying on glycolysis might lead to greater levels of fatigue, ultimately enhancing the anabolic environment [31].

In this study, we chose to use elastic bands as a practical means to maintain consistent tension on the working muscles while maximizing metabolic stress responses; factors proposed to mediate hypertrophic adaptations [32]. Low-load, short-rest RT combined with Kaatsu (blood flow restriction) and performed with elastic bands has been shown to trigger significant muscle activation [37] and hypertrophy [38]. Moreover, physiological benefits similar to weight machine RT have been observed with elastic band RT [6, 10, 16]. Elastic bands are manufactured in several strengths and can be adjusted by shortening the length of the bands to achieve the appropriate resistance for each individual’s training protocol. Furthermore, elastic bands keep continuous tension on the target muscle, making them a viable modality to achieve muscle hypertrophy. Research shows that exercise intensity can be monitored and equalized to weight machine RT using the same targeted number of repetitions and rating of perceived exertion (RPE) in active muscle [10]. Indeed, there is evidence that RT performed with elastic bands can provide an appropriate high-level stimulus to promote muscular adaptations in an athletic population [2, 21].

The purpose of the present study was to investigate if low load superset RT with the use of elastic bands enhances muscular strength, hypertrophy and endurance as compared to traditional RT. We hypothesized that the improved metabolic stress [32] and fatigue [31] experienced in superset RT performed with elastic bands would trigger anabolic factors ultimately leading to enhanced muscle cross-sectional area (CSA). Moreover, we speculated that the greater fatigue would enhance improvements in muscular endurance.

Materials and Methods

Subjects

Twenty-five young athletes (men $n = 13$, women $n = 12$) with similar daily activities and food intake were recruited from a university gymnastics club to participate in the study. Participants were stratified based on sex and then allocated via simple randomization (blind pick of a folded piece of paper with the number of a participant written on it) to one of two parallel groups: a superset group (S) or a traditional training group (T). Eligibility criteria required participants to be between 18 and 23 years old and not involved in resistance training for at least 6 months prior to the study’s onset. The composition of males and females

was chosen to provide a broad representation of the training population. Simple randomization was executed in a manner designed to obtain the same percentage of men and women in each group.

The experiment was conducted during summer break during which time the participants refrained from any structured exercise other than that performed in the study. All participants were informed about the potential risks of the experiment and gave their written consent to participate in the experiment. This study was approved by the Ethics Committee of the Nippon Sport Science University (016-H040) and was carried out in accordance with the international standards of the guidelines of the Declaration of Helsinki for Human Research (Harriss and Atkinson 2015). All data were collected at the Nippon Sport Science University, Setagaya campus, Tokyo. The sample size was calculated (GPower 3.1, Dusseldorf, Germany) a priori as follows: Effect size $f=0.7$, α err prob = 0.05, power = 0.8. The required total sample size was estimated to be $n=20$ ($n=10$ for each group).

After 2 familiarization training sessions conducted during the 2 weeks prior to study begin, both groups performed 1 exercise for the elbow flexors (standing biceps curls) and one exercise for the elbow extensors (standing overhead triceps extensions) at a resistance equating to 50%–60% RM 3 times per week for a period of 8 weeks, since this training frequency and time period has been shown sufficient to produce significant muscle hypertrophy [12]. The S group performed 1 set of biceps curls followed by 1 set of overhead triceps extensions without rest. A total of 3 sets (combination of 2 exercises) were executed with 60 s rest between sets in the S group. The T group performed 3 sets of biceps curls followed by 3 sets of overhead triceps extensions with 60 s rest between sets and exercises. Both groups used elastic bands (Therabands tubing with soft grip handles; Hygenic Corporation; Akron, OH, USA) of different levels of resistance adjusted for the 50% of 1RM for each participant by using the target number of repetitions (30–40) and the RPE in active muscle [10]; that is if the perceived exertion was lower than 6, we increased the resistance level of the band. When performing the exercises, participants stepped on the center part of the elastic band to stabilize it and executed the prescribed movement for each exercise at a cadence of 1 s on the concentric and 2 s on the eccentric phase. Biceps curls were performed by flexing the elbow from a fully extended position to the point where the contraction was maximal at about chin height. Triceps extensions were performed by extending the elbow from a fully flexed to a fully extended position behind the neck. Both groups performed as many repetitions as possible with each exercise until reaching muscular failure.

Measurements

To assess acute responses to a single bout of resistance training with regard to mechanical and metabolic stress, we measured acute muscle thickness changes, blood lactate and perceived exertion. In an effort to accustom the participants to the training protocol before acute testing, all acute measurements were conducted during one single RT session 2 weeks after the start of the training program. To assess long-term adaptations, we measured total training volume, muscle CSA, MVC, 1RM and endurance before and after the intervention.

Muscle Thickness

Acute changes of muscle thickness [20] were assessed during one RT session before and immediately after a single bout of RT via ultrasound imaging (Prosound 2; Hitachi Aloka Medical, Ltd., Tokyo, Japan). For measurements, participants were seated with their elbow extended and relaxed. Three images on the anterior and posterior surfaces measured 60% distal from the acromion process of the scapula to the lateral epicondyle of the humerus [23, 34] at the center of the arm were recorded for each participant before and immediately after RT. After application of transmission gel to the ultrasound transducer, the probe (7.5 MHz) was positioned perpendicular to the muscle without depressing the skin. The distance between the subcutaneous adipose tissue-muscle interface to the muscle-bone interface was measured via imageJ (National Institutes of Health) and the mean value of the three images was recorded as the final value. The test–retest intraclass correlation coefficient (ICC) assessed prior to the study was 0.87.

Blood Lactate

Blood lactate (BL) concentrations were measured from capillary finger blood samples collected from the finger tips using a lancet (Sanwa Kagaku Kenkyusho Co. LTD., Nagoya, Japan) during one RT session. Values before (B), immediately after (P0) and 5 min after (P5) RT were recorded by using a portable lactate analyzer (Lactate Pro 2; ARKRAY, Inc., Kyoto, Japan).

Rating of Perceived Exertion

Each participant assessed the intensity of RT based on the RPE revised category-ratio scale (0–10 scale), which is used to assess physiological and perceived stress in RT [25]. After

the last set, the 0–10 scale was shown to each participant who then assessed the RPE.

Total Training Volume

The number of repetitions and load were recorded for each set and RT session. Volume was calculated as load (% 1RM) $\times$ repetitions.

Muscle CSA

Participants underwent magnetic resonance imaging (MRI) (AIRIS II, Hitachi, Ltd., Tokyo, Japan) scans of the right upper elbow muscles (biceps, brachialis and triceps) during the week prior to the start of training and 72–96 h after the last RT session (week 9) in order to minimize the influence of acute muscle swelling on the CSA results. In order to obtain accurate measurements, markers filled with water were placed exactly at half-distance of each participant's upper elbow (measured from the lateral epicondyle of the humerus to the acromion process of the scapula). The following parameters were used to acquire 20 axial scans: repetition time/echo time, 460 ms/26 ms; field of view 20 cm, phase/frequency, 320; slice thickness, 3 mm; gap, 10 mm. The images displaying the markers were then analyzed via image (National Institutes of Health) and the square areas of the biceps and triceps muscles were calculated twice by the same investigator. The mean value was used for calculations. A reliability test showed an ICC of > 0.9 for our CSA calculations.

Maximal Voluntary Isometric Contraction (MVC)

MVC for the elbow flexor and extensor muscles were measured during the week before training began and 72–96 h after the last RT session (week 9) (Biodex System 3 dynamometer; Sakai Medical Instrument, Tokyo, Japan). While sitting in a chair, the participant's right arm was strapped at an elbow joint angle of 90° to a fixed platform at chest height. The participants held the Biodex handle in a supinated position. Each participant performed 2 MVC's separated by 60 s rest intervals for each muscle beginning with the elbow flexor. The highest value was recorded for each participant. ICC was > 0.9 for MVC measurements.

One Repetition Maximum

One repetition maximum (1RM) tests for the elbow extensors (close grip bench press) and elbow flexors (barbell curl) exercises were conducted the week before and after the training period under the supervision of qualified trainers. 1RM was measured within 5 trials with 3 min rest between each trial after 2 warm-up sets (50% 1RM $\times$ 5 reps, 60%–80%

1RM $\times$ 2–3 reps). The close grip bench press 1 RM attempt was considered successful when the barbell was in a full lock-out position with head, upper back and buttocks on the bench and both feet flat on the floor [34]. The same method was used for the barbell curl 1RM. The barbell was held in a relaxed position at shoulder-width with the elbows completely extended. The attempt was considered successful if the participant could lift the barbell from an extended position of the elbow to a fully contracted position without swinging or using momentum. ICC was > 0.9 for both 1 RM measurements.

Muscular Endurance

The day after 1RM testing, muscular endurance was measured using 50% of 1RM [34] for the close grip bench press and barbell curl performed to volitional failure. The loads for the endurance trials at the beginning and the end of the experiment were calculated from the baseline 1RM and the final 1 RM at the end of the study, respectively. Endurance tests were conducted after strength tests in order to avoid the effects of fatigue on strength results.

Statistical Methods

Data are displayed as mean $\pm$ SD. A two-way repeated measures analysis of variance (ANOVA) (time $\times$ groups) was used to analyze the data (SPSS for Macintosh version 22). Eta squared (η^2) values are reported to reflect the magnitude of the differences in each outcome. The significance level was set a priori at $P < 0.05$. Confidence intervals (CI) are reported at the 95% level.

Results

Prior to the study's completion, two female subjects dropped out for personal reasons, leaving a total of 23 subjects with the following descriptive characteristics: S (men: $n = 6$; average age: 19.2 ± 1.8 years; average height: 164 ± 6 cm; average weight: 58.5 ± 7 kg; average body fat: $10.3\% \pm 3.5\%$; women: $n = 6$; average age: 20.5 ± 0.5 years; average height: 158.3 ± 2.7 cm; average weight: 51 ± 3.8 kg; average body fat: $18\% \pm 2.6\%$); T (men: $n = 7$; average age: 19.4 ± 1.1 years; average height: 170.7 ± 5.3 cm; average weight: 63.5 ± 4.7 kg; average body fat: $9\% \pm 2.3\%$; women: $n = 4$; average age: 21.3 ± 1.3 years; average height: 154 ± 2.5 cm; average weight: 50.3 ± 6.6 kg; average body fat: $19.7\% \pm 2.1\%$). There were no baseline differences observed in any of the studied outcome measures.

Total Volume

No significant differences in total training volume between groups were observed for either exercise ($P > 0.05$). See Table 1.

Acute Measurements

Muscle Thickness

Muscle thickness of the biceps significantly increased in both the T group ($P < 0.05$) and the S group ($P < 0.05$). The triceps did not show significant increases in either T ($P > 0.05$) or S ($P > 0.05$). We did not observe any between-group differences in either outcome ($P > 0.05$).

Blood Lactate

Blood lactate showed similar changes post workout in both groups without any significant differences between groups ($P > 0.05$).

Rating of Perceived Exertion

The RPE showed no between-group differences ($P > 0.05$).

Chronic Measurements

Muscle CSA

Both groups showed significant CSA increases in the biceps ($P < 0.05$) and triceps ($P < 0.05$) without any differences noted between groups ($P > 0.05$). See Table 2.

Maximal Voluntary Isometric Contraction (MVC)

MVC of the elbow extensors showed significant increases only in the T group ($P < 0.05$); values for the S group did not reach statistical significance ($P > 0.05$). Neither group demonstrated significant increases in MVC of the elbow flexors ($P > 0.05$). We did not observe any between-group differences in this outcome ($P > 0.05$). See Table 3.

Table 1 Average number of repetitions for each set

	Biceps curl			Triceps extension		
	1st set	2nd set	3rd set	1st set	2nd set	3rd set
T	56.9 ± 19.1	31.3 ± 8.8	32.3 ± 9.8	34.2 ± 14.3	22.3 ± 9.6	19.6 ± 7.9
S	60.1 ± 23.7	43.8 ± 8.1	33.9 ± 8.4	28.7 ± 7.2	21.4 ± 4.4	16.5 ± 5.4

All values are expressed as mean ± SD

Table 2 Changes in CSA before and after the training period

	Biceps		Mean difference (95% CI)	η^2	Triceps		Mean difference (95% CI)	η^2
	Pre	Post			Pre	Post		
T	2.2 ± 0.6	2.5 ± 0.7			5.6 ± 1.7	6.0 ± 1.5		
S	2.1 ± 0.5	2.3 ± 0.6	−0.02 (−0.18 to 0.13)	0.000	5.2 ± 1.0	5.4 ± 1.2	−0.18 (−0.57 to 0.21)	0.001

All values are expressed as mean ± SD. Units are arbitrary units

CI confidence interval

Table 3 Changes in maximal voluntary isometric contraction before and after the training period

	Arm flexor		Mean difference (95% CI)	η^2	Arm extensor		Mean difference (95% CI)	η^2
	MVC (N·m)				MVC (N·m)			
	Pre	Post			Pre	Post		
T	43.1 ± 10.5	46.1 ± 13.7			42.1 ± 11.5	45.8 ± 12.3		
S	37.3 ± 10.7	37.5 ± 12.1	−2.62 (−9.31 to 4.07)	0.003	39.8 ± 11.1	43.2 ± 13.8	−0.26 (−6.22 to 5.69)	0.003

All values are expressed as mean ± SD

CI confidence interval

One Repetition Maximum (1RM)

One repetition maximum for the bench press significantly increased in the T group only ($P < 0.05$); changes in the S group did not reach statistical significance ($P > 0.05$). Changes in one repetition maximum for the barbell curl did not reach statistical significance in either group. We did not observe any between-group differences in either outcome ($P > 0.05$). See Table 4.

Muscular Endurance (Table 5)

Muscular endurance significantly increased for the S group for both the close grip bench press (from 24.2 ± 8.6 reps to 30.3 ± 11.1 reps; $P < 0.05$) and the barbell curl (from 52.8 ± 24.5 reps to 61.6 ± 26.7 reps; $P < 0.05$). No statistically significant changes were observed in the T group for the bench press (from 29.9 ± 9.0 reps to 32.9 ± 9.0 reps; $P > 0.05$) or the barbell curl (from 60.3 ± 32.9 reps to 60.0 ± 36.8 reps; $P > 0.05$). We did not observe any between-group differences in either outcome ($P > 0.05$). See Table 5.

Discussion

We compared the acute effects of traditional and superset training on metabolic and mechanical stress markers as well as the effects on muscular strength, mass and endurance over an 8-week study period. The study produced several notable findings. Although no between-group differences were observed for muscular strength, only the traditional group significantly increased bench press 1RM and MVC for the elbow extensors. Alternatively, only the superset

group showed statistically significant increases in markers of muscle endurance, again without observed between-group differences. Both groups significantly increased CSA of the biceps and triceps without statistically significant differences between groups. When considered in total, these findings suggest that superset training may be beneficial to muscular endurance while attenuating strength gains; there does not seem to be a benefit or detriment to this method on muscle hypertrophy.

Research regarding intensity and mechanical-metabolic stress response in superset RT is conflicting. One previous study demonstrated that energy expenditure and lactate concentrations in superset RT are larger when compared to traditional set RT [19], whereas a recent study showed similar energy expenditure for superset and traditional RT [7]. In this study, acute stress markers such as blood lactate and RPE showed no between-group differences. However, acute muscle thickness increases showed differences for the muscle trained in the second half of the superset. Thus, speculation that superset training may enhance muscular adaptations via greater metabolic stress is unfounded, at least during performance of relatively light-load RT. The lack of significant increases in muscle thickness for the triceps in the S group might be due to the exercise order. The S group performed triceps extensions directly after performing biceps curls to failure, possibly resulting in an increased accumulation of fatigue and blood lactate, which in turn potentially hindered maximum performance for the ensuing triceps exercise.

The CSA of the biceps and triceps significantly increased in both groups, with no statistical differences observed between conditions. Thus, in opposition as to what has been previously speculated [31], combining agonist and antagonist muscle groups in superset fashion did not lead to

Table 4 Changes in one repetition maximum before and after the training period

	Bench press (kg)		Mean difference (95% CI)	η^2	Barbell curl (kg)		Mean difference (95% CI)	η^2
	Pre	Post			Pre	Post		
T	53.1 ± 21.5	56.7 ± 21.8			29.8 ± 6	30.5 ± 6.2		
S	45.2 ± 14.9	47.5 ± 16.5	$-1.31 (-6.71 \text{ to } 4.09)$	0.006	26.2 ± 7.9	27.3 ± 7.7	$0.22 (-1.20 \text{ to } 1.64)$	0.004

All values are expressed as mean $\pm$ SD

CI confidence interval

Table 5 Changes in muscular endurance (number of repetitions at 50% 1RM) before and after the training period

	Bench press (reps)		Mean difference (95% CI)	η^2	Barbell curl (reps)		Mean difference (95% CI)	η^2
	Pre	Post			Pre	Post		
T	29.9 ± 9.0	32.9 ± 9.0			60.3 ± 32.9	60.0 ± 36.8		
S	24.2 ± 8.6	30.3 ± 11.1	$2.86 (-2.90 \text{ to } 8.62)$	0.007	52.8 ± 24.5	61.6 ± 26.7	$9.53 (-0.23 \text{ to } 19.28)$	0.006

All values are expressed as mean $\pm$ SD

CI confidence interval

superior muscle hypertrophy. Even though the training load was relatively light, each set was carried out to muscular failure in both groups. Metabolic stress measured via blood lactate, RPE and acute muscle swelling did not differ among traditional and superset RT, indicating the possibility that anabolic signaling triggered via mechanical and metabolic stress remains similar for both RT protocols. These findings run contrary to the hypothesis that, by enhancing reciprocal inhibition of the antagonist, superset training elicits greater mechanical forces in the agonist that in turn promotes a corresponding heightened anabolic response [31].

Previous research conducted with low-load, short-rest RT combined with Kaatsu reported significant CSA increases (biceps: 17.6%, triceps: 17.4%) after 12 weeks [38]. In the present study, we observed robust increases in biceps CSA ($13.2\% \pm 5.0\%$ and $12.9\% \pm 7.3\%$ for traditional and superset training, respectively) and triceps CSA ($9.5\% \pm 9.3\%$ and $4.8\% \pm 4.1\%$ for traditional and superset training, respectively) after 8-week RT. Despite the fact that the training period length was 4 weeks shorter in our study, comparable CSA increases were observed between studies, indicating that low-load, short-rest RT with resistance bands performed to failure can lead to similar results as when combined with Kaatsu for a fixed number of repetitions. Indeed, the number of repetitions in this study was much higher due to taking each set to failure, probably triggering similar metabolic stress as is observed when using Kaatsu for a fixed number of repetitions not to failure. Our results emphasize the importance of training with high levels of effort with low-load RT to maximize muscle hypertrophy. Low-load RT carried out well short of failure might be suboptimal for fully activating high-threshold motor units whereas using high levels of effort would promote more complete motor unit activation, conceivably due to elevated metabolic stress [32]. The constant tension on the target muscle achieved via training with elastic bands might have contributed to the high levels of metabolic stress, which in turn may have enhanced CSA increases.

Measures of maximal strength increased only in the traditional group, as assessed by the 1RM bench press and MVC for the elbow extensors. Our results differ with some previous research showing improved strength increases for superset RT [27]. However the study differed from ours in training load and rest periods, making a direct comparison unfeasible.

Our results for the 1RM barbell curl partially reaffirmed that the use of low loads combined with short rest intervals are suboptimal for increasing strength [12]. On the other hand, MVC for the elbow extensors showed a significant pre-to post-study increase in the traditional group, albeit of a relatively modest effect. Indeed, a recent review demonstrated that training with loads $< 60\%$ 1 RM can enhance maximal strength but the magnitude of the increase is inferior

compared to high load RT ($> 60\%$ 1RM) [33]. These results suggest that traditional RT might be superior with regard to strength gains as compared to superset RT. We can only speculate that the increased rest time between muscle groups during traditional training may have allowed for improved recovery and strength adaptations. Recent studies also propose that heavy load RT might favorably stimulate type II muscle fibers whereas low load RT seems to enhance type I fibers [24], possibly explaining the discrepancies in strength gains between high and low load RT. It should be noted that the point estimate differences for strength outcomes between conditions were relatively small and the associated confidence intervals were relatively large, casting doubt on the practical meaningfulness of these findings.

Muscular endurance showed statistical increases in the S group only. Moreover, the point estimate differences between conditions and associated confidence intervals suggest the findings may be practically meaningful. To our knowledge, this is the first study investigating the effects of superset RT on muscular endurance. When performing a superset, the number of repetitions performed in one set is much larger as compared to traditional set RT. This factor conceivably leads to superior endurance adaptations with superset RT. Previous research has shown a significant effect of low-load training on muscular endurance [34]. It is not clear why the group that performed traditional low-load training did not see such enhancements. Possible reasons include differences in the modality of training (i.e. resistance bands versus free weights) and the type of exercises employed (single-joint versus multi-joint). This topic requires further study.

There are several limitations to this study. First, even though we tried to match the number of men and women in each group, the final numbers were unequal due to drop outs. Previous research on gender differences in response to RT is controversial; one study indicates that the rate of muscle hypertrophy is similar in men and women [11], whereas another study conducted on older adults found that the rate of muscle fiber growth and strength increases were superior in men as compared to women [5]. Second, the training period of 8 weeks might not have been optimal to assess the extent of physiological adaptations. However, a previous study showed that the biceps and triceps CSA increased after 6-week RT with no significant improvements after 8 and 12 weeks as compared to 6 weeks [1]. Third, the close grip bench press exercise is a multi-joint exercise including the triceps but this exercise also stresses the pectorals and shoulder muscles to a certain extent. However, in order to measure 1RM for the elbow extensors, the close grip bench press is more suitable as compared to single-joint exercises such as the French press for safety reasons. Indeed, 1RM attempts on the French press potentially can lead to serious elbow joint injuries. Fourth, this study was conducted without active/passive control groups. Finally, the target effect

size for calculating sample size was relatively large; thus, the study may have been underpowered to detect smaller changes that may be practically meaningful.

Practical Applications

Compared to traditional RT, superset RT displays similar muscle hypertrophy adaptations. Our findings suggest that superset training may produce superior improvements in muscular endurance, at least when it is employed with relatively low loads. Therefore, the method conceivably can be used for male and female trainees aiming to increase muscular endurance while gaining muscle mass. Moreover, superset RT is very time efficient, making it an attractive training method for trainees with time restrictions with regard not only to endurance, but also to enhance muscular development. However, strength outcomes might be inferior to traditional RT and therefore superset RT might not be the best option for trainees aiming to maximize force-producing capacity.

Conclusions

In conclusion, low-load superset RT performed to failure using relatively light loads might produce superior improvements in muscular endurance compared to traditional training; however, the superset method seems to elicit inferior increases in strength while having a neutral effect on muscle hypertrophy. Training to failure with a relatively low load probably triggers high levels of mechanical and metabolic stress leading to the physiological adaptations mentioned above. However, the decreased time to recover between muscle groups might impair strength adaptations in superset RT.

Compliance with Ethical Standards

Conflict of interest The authors have no conflicts of interest. The results of the current study do not constitute endorsement of the product by the authors or the journal.

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